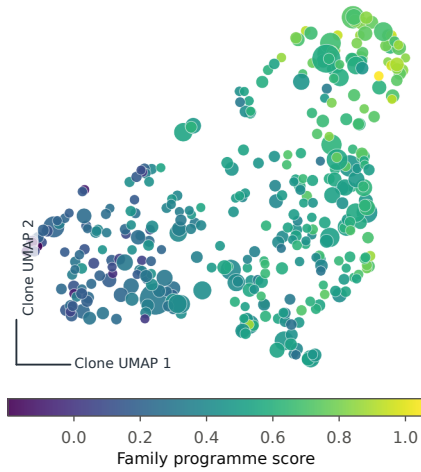


Supplementary Figure 11 | RBD clone map: measured-anchored zone axis and parameter robustness

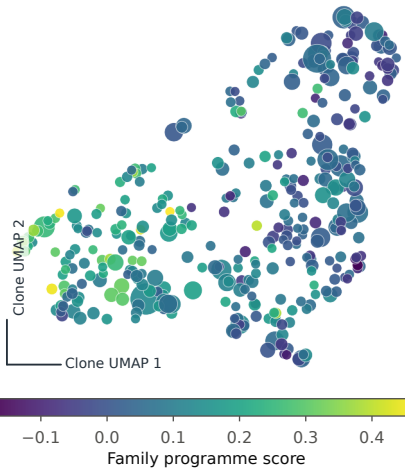
Programme scores reuse expression and are descriptive; sort gates and division history are measured.

A Dark-zone / cycling programme score



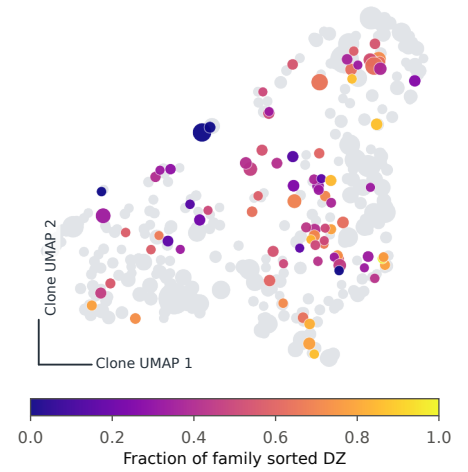
381 reliable clones; clone UMAP

B Light-zone programme score (incl. Myc)



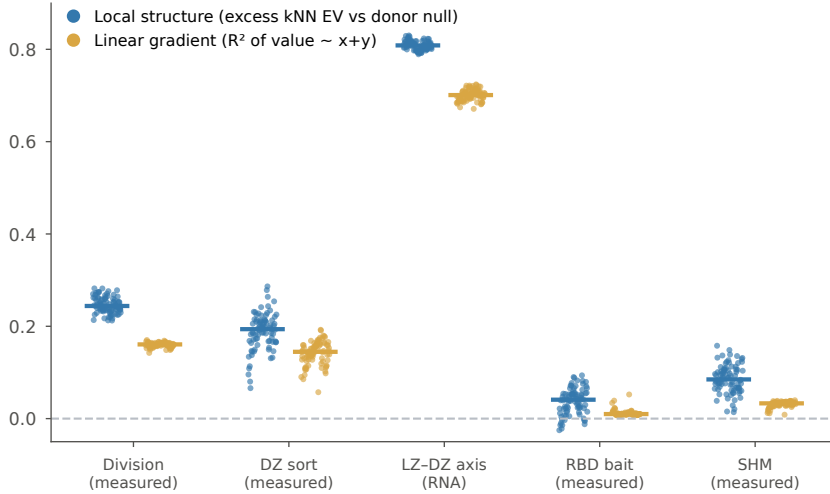
381 reliable clones; clone UMAP

C Measured DZ sort fraction (mRNA arm only)



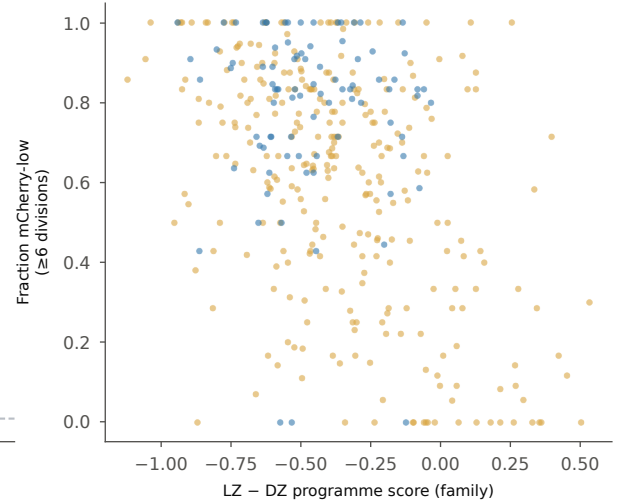
Grey: all 381 reliable clones; colour: 87 clones with zone-gated cells (5 mRNA-arm mice).

D Map associations across 85 embedding parameter sets



85 UMAP parameter sets ($n_{\text{neighbors}}/\text{min_dist}/\text{spread}/\text{seed}$), map rebuilt from committed family features each time; bars = median. SHM: linear $R^2 \leq 0.04$ in all sets.

E The measured division gradient tracks the DZ-LZ programme axis



$r = -0.44$; donor-stratified permutation $p = 0.005$; $n = 381$ families.